ODE shapes to recognise on sight

0 · Which method? (look at the coefficients)

You see	Do
constant coefficients	$e^{\lambda t}$ (§2)
every term $t^m y^{(m)}$	Euler, t^n (§3)
Euler $+\,k^2r^2R$	Bessel, J_0, Y_0 (§4)
$y'' - ty = 0$	Airy: series or FT (§5)
polynomial coefficients	power series (§6)
one solution v given	reduction of order (on sheet)
const. coeff. $+\ \delta$ / H / piecewise, ICs at 0	Laplace transform
$x \in \mathbb{R}$, $x\,y$ term or δ forcing	Fourier transform (§10)
forcing given as a Fourier series	term by term (§9)

1 · First order

Spot: $y' + f(t)\,y = g(t)$ (divide by the y' coefficient first).

$$I = e^{\int f\,dt}, \qquad (Iy)' = Ig, \qquad y = \frac{1}{I}\Big(\int Ig\,dt + C\Big)$$

Quick I: $f = \frac{a}{t} \to t^a$; $f = 2t \to e^{t^2}$; $f = \tan t \to \sec t$.

Spot: $y' = f(t)\,g(y)$ (separable): $\displaystyle\int \frac{dy}{g(y)} = \int f\,dt$.

Spot: $y' = ky$: $y = Ce^{kt}$. Heat time ODE $\dot T = -\alpha k^2 T \Rightarrow T = e^{-\alpha k^2 t}$.

2 · Constant coefficients

Spot: $ay'' + by' + cy = 0$. Trial $e^{\lambda t}$: $a\lambda^2 + b\lambda + c = 0$.

$$y = \begin{cases} C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}, & \lambda_1 \neq \lambda_2 \text{ real} \\ (C_1 + C_2 t)\,e^{\lambda t}, & \text{repeated} \\ e^{\alpha t}(C_1 \cos\beta t + C_2 \sin\beta t), & \alpha \pm i\beta \end{cases}$$

Separation-of-variables ODEs, write instantly:

$$\begin{aligned} X'' + k^2 X = 0 &\ \Rightarrow\ A\cos kx + B\sin kx \\ X'' - k^2 X = 0 &\ \Rightarrow\ A\cosh kx + B\sinh kx \\ X'' = 0 &\ \Rightarrow\ Ax + B \\ \ddot T + c^2k^2 T = 0 &\ \Rightarrow\ A\cos ckt + B\sin ckt \\ \dot T + \alpha k^2 T = 0 &\ \Rightarrow\ Ce^{-\alpha k^2 t} \end{aligned}$$

cosh / sinh on a finite interval; zero at $y = b \Rightarrow \sinh k(b-y)$. On $[0,\infty)$ with boundedness $\Rightarrow e^{-kx}$ only.

3 · Euler (Cauchy–Euler)

Spot: $at^2y'' + bty' + cy = 0$: power of t = order of derivative, every term.

$$y = t^n \ \Rightarrow\ an(n-1) + bn + c = 0$$

$$y = \begin{cases} C_1 t^{n_1} + C_2 t^{n_2}, & n_1 \neq n_2 \text{ real} \\ t^n(C_1 + C_2 \ln t), & \text{repeated} \\ t^\alpha\big(C_1\cos(\beta\ln t) + C_2\sin(\beta\ln t)\big), & \alpha \pm i\beta \end{cases}$$

Polar Laplace ($\Theta = \cos n\theta, \sin n\theta$):

$$\begin{aligned} r^2R'' + rR' - n^2R = 0 &\ \Rightarrow\ Ar^n + Br^{-n} \quad (n \geq 1) \\ r^2R'' + rR' = 0 \quad (n=0) &\ \Rightarrow\ A + B\ln r \end{aligned}$$

Centre in domain $\Rightarrow B = 0$. Forcing $t^k \to At^k$; resonance $\times(\ln t)^s$.

4 · Bessel, order 0

Spot: Euler shape plus an extra k^2r^2R (or $R'' + \frac{1}{r}R' + k^2R = 0$).

$$r^2R'' + rR' + k^2r^2R = 0$$
$$R = A\,J_0(kr) + B\,Y_0(kr)$$

$$J_0(s) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2}\Big(\frac{s}{2}\Big)^{2m}, \qquad J_0(0) = 1$$

$Y_0 \sim \ln r$ at $0 \Rightarrow B = 0$ if the centre is in the domain.
$R(a) = 0 \Rightarrow J_0(ka) = 0 \Rightarrow k_n = \frac{z_n}{a}$, z_n the zeros of J_0 ($z_1 \approx 2.40$).
Comes from heat / wave on a disc (time present). Order n: $(k^2r^2 - n^2)R \to J_n, Y_n$ (recognise only).

5 · Airy

Spot: $y'' - ty = 0$ (no y', coefficient t).

Series: $t\,y$ shifts by 3, so $c_2 = 0$ is forced:

$$c_{m+3} = \frac{c_m}{(m+3)(m+2)}$$

Two chains, t^{3k} (from c_0) and t^{3k+1} (from c_1); $R = \infty$.
FT version (sheet rules $(ik)^2\hat y$, $i\hat y'$):

$$-k^2\hat y - i\hat y' = 0 \ \Rightarrow\ \hat y = Ae^{ik^3/3} \ \Rightarrow\ y = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ik^3/3 + ikx}\,dk$$

6 · Power-series families

Spot: $(a + \alpha t^2)y'' + bty' + cy = 0$.

$$c_{m+2} = -\frac{\alpha m(m-1) + bm + c}{a(m+2)(m+1)}\,c_m$$

Polynomial solution: numerator $= 0$ at $m = k$. Radius $\geq$ distance to nearest complex zero of the y'' coefficient; here $R = \sqrt{|a/\alpha|}$.
Spot: $p\,t\,y'' + q\,y' + \ldots$ (singular at 0): only c_0 free, one series solution.

$$\mathrm{wr} = C\,t^{-q/p}, \qquad y_2 = y_1\int \frac{\mathrm{wr}}{y_1^2}\,dt$$

$q/p = 1 \Rightarrow y_2 \sim \ln t$. Indicial ($t^\lambda$ trial): $\lambda(p\lambda - p + q) = 0$.

7 · Wronskian / Abel

Standard form $y'' + f_1y' + f_0y = 0$:

$$\mathrm{wr} = y_1y_2' - y_1'y_2$$
$$\mathrm{wr}(t) = \mathrm{wr}(t_0)\,e^{-\int_{t_0}^t f_1}$$
$$\Big(\frac{y_2}{y_1}\Big)' = \frac{\mathrm{wr}}{y_1^2}$$

For two solutions, wr is never 0 or always 0.

8 · Systems $\mathbf{y}' = A\mathbf{y}$

$$\lambda^2 - (\mathrm{tr}\,A)\lambda + \det A = 0$$
$$\mathbf{y} = C_1\mathbf{v}_1e^{\lambda_1 t} + C_2\mathbf{v}_2e^{\lambda_2 t}$$

$y'' + by' + cy = 0 \leftrightarrow A = \begin{pmatrix} 0 & 1 \\ -c & -b \end{pmatrix}$, $\mathbf{y} = (y, y')$.

Eigenvector: row (p, q) of $A - \lambda I \Rightarrow \mathbf{v} = (q, -p)$.
$\det < 0$ saddle. $\det > 0$: $\mathrm{tr}^2 > 4\det$ node, $\mathrm{tr}^2 < 4\det$ spiral, $\mathrm{tr} = 0$ centre. $\mathrm{tr} < 0$ stable.

9 · Particular solutions

Forcing	Trial
e^{kt}	Ae^{kt}
degree-m polynomial	full degree-m polynomial
$\cos\omega t$ or $\sin\omega t$	$A\cos\omega t + B\sin\omega t$
Euler with t^k	At^k

Resonance: trial already solves the homogeneous ODE (s-fold root) $\Rightarrow \times t^s$ (Euler: $\times(\ln t)^s$).

$$y'' + \omega^2 y = \cos\omega t \ \Rightarrow\ y_p = \frac{t}{2\omega}\sin\omega t$$

Fourier forcing, $s(x) = \sum s_n \cos nx$:

$$y'' - ky = s(x)$$
$$y = \sum_n \frac{-s_n}{n^2 + k}\cos nx$$

Steady state with source: $u_S'' = -q/\alpha$, integrate twice.

10 · Fourier-transform ODEs

Pair (not on sheet): $e^{-a|x|} \leftrightarrow \dfrac{2a}{a^2 + k^2}$.

$$y'' - a^2y = -\delta(x) \ \Rightarrow\ y = \frac{e^{-a|x|}}{2a}$$

$y^{(n)} \pm x\,y = 0$: the $x\,y$ term becomes $i\hat y'$, so you get a first-order ODE in k. Solve it with an integrating factor, then leave y as an inverse-FT integral.

Not in this year's lectures

Legendre, Hermite, Bernoulli, full Frobenius. If one appears, the question will give the method.